

Section - ASubsential

- 1) Budding ... ?
- 2) Hardy - Weinberg's Principle
- 3) Self-pollination
- 4) Progesterone
- 5) ICSE - Intracytoplasmic sperm injection
- 6) Natural passive immunity
- 7) Methanobacterium
- 8) Convergent evolution
- 9) Turner's Syndrome
- 10) Linkage
- 11) BOD (Biochemical Oxygen Demand)
- 12) Helium
- 13) Height
- 14) ~~Corpus~~ Corpus luteum

- 15) Growth Inhibition
- 16) Amensalism
- 17) Integrates the viral double-stranded DNA into the host's cell's chromosome.
- 18) Stationary phase
- 19) Zona pellucida
- 20) Rheumatoid arthritis

Section-B

- | | |
|--|---|
| <p>21) <u>ZIFT</u></p> <p>→ Zygote Intra-fallopian transfer</p> <p>→ Fertilisation occurs outside the body & is then transfused into the fallopian tube.</p> | <p><u>GIFT</u></p> <p>→ Gamete Intra-fallopian transfer.</p> <p>→ An unfertilised ovum which was collected from a donor is transfused into the fallopian tube along with sperm of the female.</p> |
|--|---|

- 22) When a single gene influences 2 or more phenotypic traits, it's called pleiotropy.
 Eg: PKU (Phenylketonuria) in humans

23) They are located outside the abdominal cavity because the scrotum maintains a temperature of $2-3^{\circ}\text{C}$ lower than the internal body temperature, which is required for proper motility and growth of the sperm.

~~Sertoli~~ Sertoli cells, also known as nurse cells, provide nutrition and secrete inhibin, for the process of spermiogenesis.

24) Primary Secondary

- Physical process involving filtration & sedimentation to remove large and small debris.
- Biological process where flocs in aeration tanks consume dissolved organic matter, and reduce the BOD.

25) • Hibernation (winter sleep): Organisms avoid the extremely cold winter by drastically reducing metabolic activity and falling into a deep sleep.

Eg: Bears

• Aestivation (Summer sleep): Organisms avoid the extremely hot summer by deep summer sleep.

Eg: Snails

26) • It helps to terminate pests by toxin activation.

27) Primary Secondary

- Sites where immature lymphocytes originate & bind with antigens to generate mature into antigen-sensitive lymphocytes.
- Sites where mature lymphocytes bind with antigens to generate an effector immune response.

Eg: Spleen

9

Section - C

28) **Exine:** Hard & durable outer layer made up of sporopollenin, containing germ pores where sporopollenin is absent.

Intine: Inner, thin & continuous wall layer made of cellulose.

Vegetative cell: Large cell containing abundant food reserves & contains an irregular-shaped nucleus that drives pollen tube growth.

Generative cell: Small spindle-shaped cell with a dense cytoplasm that is present in the cytoplasm of the vegetative cell & divides mitotically to produce 2 male gametes.

29) **Females (Queen/workers):** Develop from fertilised eggs ($2n$) and are diploid.

Males (Drones): Develop from unfertilised eggs by parthenogenesis (n) and are haploid.

30) Red-green colourblindness is an X-linked recessive trait (X^c)

Father: $X^C Y$

Mother (carrier): $X^C X^c$

X^C	X^C	X^c
X^c	$X^C X^C$	$X^C X^c$
Y	$X^C Y$	$X^c Y$

$X^C X^C$ = Normal daughter

$X^C X^c$ = Carrier daughter

$X^C Y$ = Normal son

$X^c Y$ = colourblind son

3)

Chance of having colourblind daughters: 0%
Chance

31) Nuclear endosperm: Liquid endosperm

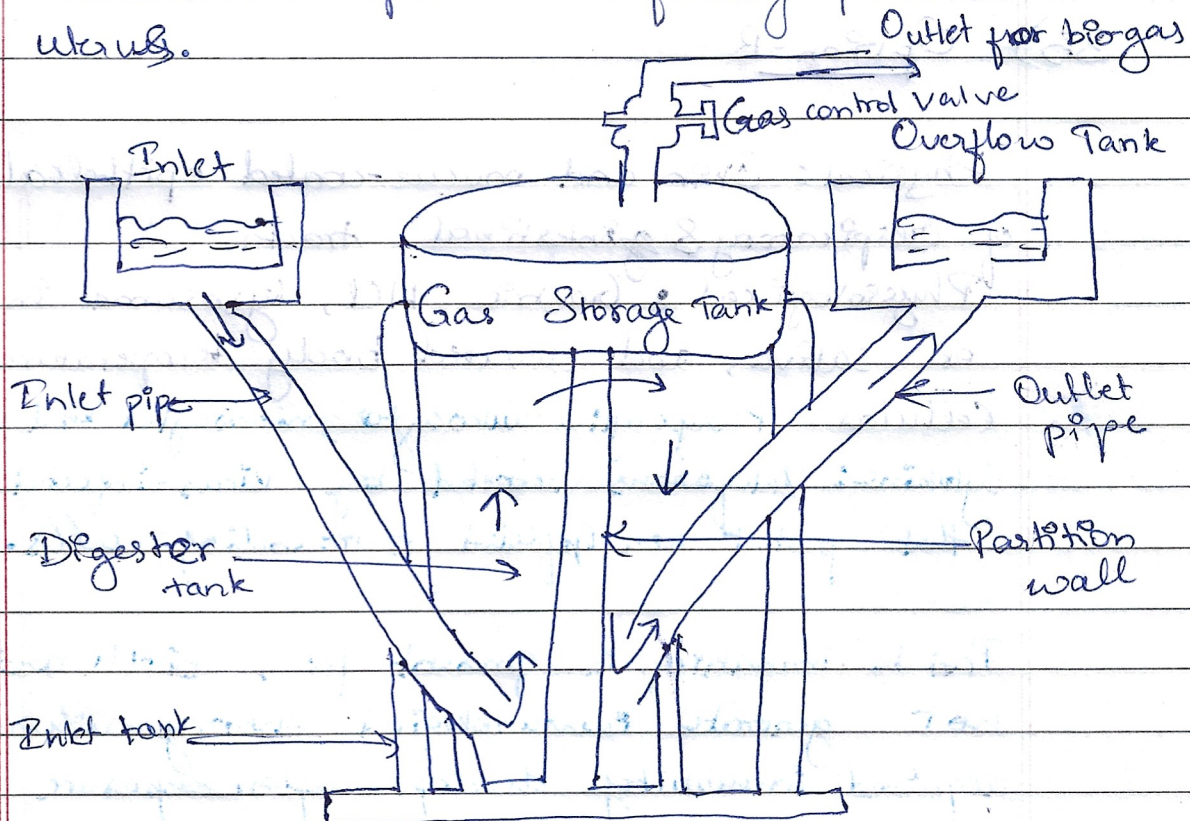
Eg: Coconut water

Cellular endosperm: -

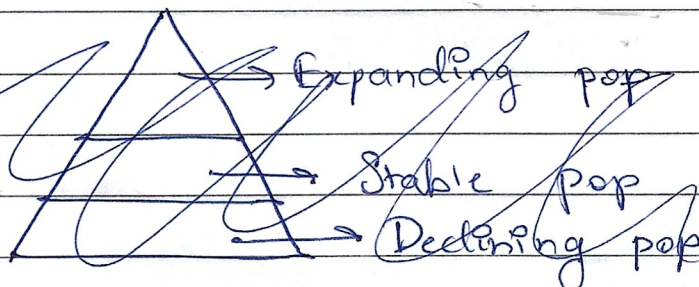
32) AI: It is a technique where semen is collected from the male and is artificially injected into the female's vagina.

IUI: It is a form of AI where washed, concentrated sperm is artificially placed in the uterus.

33)



34)



34) Expanding pop: It is triangular. Broad base with a high proportion of pre-reproductive individuals. Birth rate exceeds death rate.

Stable pop (Bell-shaped): Pre-reproductive & reproductive age groups almost equal, causing near 0 pop. growth.

Declining pop (Urn-shaped): Narrow base - low birth rates, pre-repro. group is smaller than reproductive group.

Section-D

35) Option-B

Physical: Skin and mucus-coated epithelial linings of respiratory & gastrointestinal tracts.

Physiological: Gastric HCl, lysozyme in tears or saliva, and elevated body temperature (fever).

Cellular: Phagocytic leukocytes, monocytes and NK cells.

Cytokine: Interferons secreted by virus-infected cells that protect non-infected surrounding cells.

Innate immunity is present from birth and does not generate immunological memory whereas acquired immunity develops upon exposure and forms memory cells.

36) The evolutionary process where an older species diversifies into multiple diverse forms occupying different ecological niches in the given area is adaptive radiation.

Darwin's Finches (Galapagos Islands): From an original seed-eating ancestral finch on the mainland, diverse beaks evolved to exploit different food niches.

Australian Marsupials: Starting from an ancestral stock, a variety of marsupials evolved within the isolated Australian continent, each adapted to different habitats.

37) (i) Mutualism: An interaction where both species are affected positively. (+/+)

Eg: Lichens (Fungus and Algae)

(+/-) (ii) Parasitism: A interaction where 1 species derives nutrition from the other species, leading to the benefit of the first and loss of the second. Eg: Mosquito (parasite) and human (host)

(iii) Predation: An interaction where the predator captures, kills and feeds on the prey. (+/-)

(iv) Commensalism: An interaction where 1 organism is benefitted positively while the other is not affected at all. Eg: Barnacles on a whale (+/0)

(v) Competition: An interaction where both organisms compete for limited resources.

Eg: Abingdon tortoises and goats on Galapagos Islands, leading to the extinction of the tortoises (-/-).